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ORIGINAL ARTICLE

EXERCISE PHYSIOLOGY AND BIOMECHANICS

Effects of horizontally *versus* vertically loaded resistance exercises on performance and muscle architecture

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ABSTRACT

BACKGROUND: The aim of this study was to compare a training program based on horizontally (HT) *versus* vertically (VT) loaded exercises on performance and muscle architecture of the lower body muscles.

METHODS: Nineteen resistance trained individuals were randomly assigned to HT (N.=10; age: 25.9±4.2 y; body mass: 72.7±11.4 kg; height: 174.0±6.0 cm) or VT group (N.=9; age: 26.9±4.4 y; body mass: 76.2±10.8 kg; height: 174.2.0±5.8 cm). Both 6-week training programs included 4 training sessions per week and were equated for the total number of repetitions. One repetition maximum (1RM) was assessed for squat and hip thrust, together with vertical and horizontal jumps and sprint. Muscle thickness (MT) and echo intensity (EI) of vastus lateralis, vastus medialis and gluteus were also evaluated pre- and post-training period.

RESULTS: A significantly greater increase in 1RM hip thrust was detected in HT (+17.9%; P=0.004) while greater increases in 1RM squat were found in VT (+10.5%; P=0.007). A greater increase (P=0.009) in vastus medialis MT was detected in VT (4.1%) compared to HT (-7.9%). Similar increases in MT of gluteus were registered in both groups (P<0.05). A greater improvement in standing long jump (P=0.004) was detected in HT (+7.6%) compared to VT (+1.6%), while both groups significantly improved vertical jump performance. Combining both groups, strong correlations were detected between gluteus EI and 20-m sprint (r=0.79; P<0.001).

CONCLUSIONS: Results indicate that HT was more effective than VT for horizontal jumps while both HT and VT were equally effective on vertical jumps. Both HT and VT promoted similar changes in muscle architecture of the gluteus, but not of the vastus medialis.

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KEY WORDS: Muscles; Exercise; Sports.

Lower body resistance exercises are widely used by athletes and gym goers to improve legs power, strength, and muscle size.[1] Some resistance exercises involving the hip extensors muscles are characterized by an antero-posterior vector relative to the human body and are defined as horizontally loaded.[2, 3] The hip thrust and the reverse hyperextension, exercises that activate the muscles of the posterior chain (*i.e.* gluteus muscles), are included in this category.

Conversely, other resistance exercises are characterized by an axial force vector and are considered vertically loaded (*i.e.* squat and step-up). Both horizontally and vertically loaded exercises may induce changes in sprint and jumping performance, and stimulating muscle growth in the lower body muscles. Some authors however,[3] hypothesized that the inclusion of vertically loaded exercises into a training program may have major effects on vertical jump while the inclusion of the horizontally loaded exercises may be more effective in improving sprint performance. Most of the training studies comparing lower body exercises with different vectors however, included the hip thrust and the squat exercises only, within their training program. Resistance training programs usually include several exercises for the lower body muscles, and resistance exercises and jumps are often combined into complex routines.[4-6]

The reverse hyperextension exercise was developed to

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target the muscles of the posterior chain in competitive athletes[7] and some authors reported greater excitation of these muscles compared to the hyperextension.[8] Although the growing popularity of this exercise, to date, no training interventions investigated the inclusion of this movement into a lower body training program. Step-up is a vertically loaded multijoint exercise performed with one leg at a time that is often prescribed to strengthen the hip and knee extensors in athletes and practitioners.[9] Despite several studies investigated muscle excitation during this exercise,[10] to the best of our knowledge, no studies were conducted on the chronic effect of this exercise alone or in combination with other lower body resistance exercises (*i.e.*, parallel squat). In addition, studies that compared the effects of vertically *vs.* horizontally loaded exercises were conducted on young individuals (14-17 years old). Contreras and colleagues[11] compared the effects of a 6-week training program including either hip thrust or front squat. The hip thrust group improved more in sprinting and mid tight pulling force compared to the squat group, which improved more in vertical jumping performance. In another interesting 20-week training study, conducted by Abade *et al.*[6] on young soccer players, vertically loaded exercises improved vertical jump, horizontal jump, and sprint performance, while horizontally loaded exercises showed improvements in horizontal jump, and sprint performance. Back-squat and hip-thrust showed an important transfer effect to both jumping and sprinting performance. These authors, however, did not consider changes in muscle architecture that may occurred following a training period based on either vertically loaded or horizontally loaded exercises. Recently, a systematic review and meta-analysis on adult basketball players reported that vertically loaded exercises were more effective than horizontally loaded exercises to enhance the change of direction capability and concluded that mixed vertical and horizontally loaded exercises may be the most appropriate solutions.[12]

Thus, the aim of the present study was to compare the effect of a lower-body training program based on horizontally loaded exercises to a training program including vertically loaded exercises, on sprint, jump, maximal strength performances, and on quadriceps and gluteus muscles architecture in trained individuals. It was hypothesized that the HT would improve sprint, jump, maximal strength performances and the gluteus muscle thickness to a greater extent than VT because of specificity. On the contrary, it was hypothesized that VT would elicit greater gains in muscle thickness of the quadriceps compared to HT in trained adults.

Materials and methods

Experimental approach to the problem

Participants were randomly assigned to either HT or VT groups. Both groups followed a 6-week training program using the same exercises for the upper body and different exercises for the lower body (horizontally or vertically loaded exercise in VT and HT, respectively). The two programs were equated for the total number of repetitions provided. Subjects were assessed for muscle architecture, body composition, strength, power, and sprint performance before (pre) and after (post) the training period. Post tests were performed 72 hours following the last training session. Based on previous data,[13] the estimated sample size was 11 in each group, to detect a between-group difference of 7.2 and 12.1 kg in the one repetition maximum (1RM) bench press and squat, respectively, with a power of 0.80. This analysis was performed using the G*POWER 3 software (G-Power, Brunsbüttel, Germany).

Participants

Nineteen resistance-trained individuals who trained for a minimum of 3 times per week in the 2 years previous to the study (mean experience in resistance training: 7.6±6.0 years) participated in the present study. Inclusion criteria required participants to squat at least their body mass (mean 1RM squat: 109.27±35.42 kg) for both groups. Participants, 15 men and 4 women (2 in each group), were randomly assigned to the VT group (N.=9; age: 26.9±4.4; body mass: 76.2±10.8 kg; height: 174.2.0±5.8 cm, 1RM squat/body mass: 1.54±0.47, 1RM hip thrust/body mass: 1.67±0.37) or to the HT group (N.=10; age: 25.1±4.0 y; body mass: 70.5±11.7 kg; height: 173.8±6.2 cm, 1RM squat/body mass: 1.48±0.42, 1RM hip thrust/body mass: 1.83±0.46). All participants signed an informed consent document after being informed about the potential risks. The study was approved by the local bioethical review board (Protocol N. 0025317 of 2/2/2023).

Resistance training programs

The exercises included in both VT and HT resistance training programs and the progression of loads in the 6 weeks of training are reported in Table I, II, respectively. All participants exercised 4 days per week for 6 weeks. Both groups included the same upper body resistance exercises and differed for the lower body exercises only. The VT program included parallel squat, barbell step-up and vertical jumps (countermovement jumps landing on a 30

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TABLE I.—*Exercises included in the VT and in the HT resistance training program.*

Parameters	Day			
	Day1	Day2	Day3	Day4
Vertically loaded exercises	Parallel squat Barbell step up	Box jumps	Parallel squat Barbell step up	
Horizontally loaded exercises	Hip thrust Reverse hyperextension	Standing long jumps	Hip thrust Reverse hyperextension	
Common exercises for both groups	Seal row Lat machine Cable pulley row Scott biceps curl Standing biceps curl	Bench press Inclined bench press Cable cross Skull crusher Triceps push down	Military press Lateral raises Front raises Dumbbell shoulder press	Sprints

HT: horizontal; VT: vertical; VL: vastus lateralis muscle.

TABLE II.—*Progression of loads and sprints in the 6 weeks of training of both VT and HT groups.*

Weeks	Strength workout/sprint
1	Lower body: 5 sets of 8 reps, RIR1, rest 120 s Upper body: 5 sets of 5 reps, RIR2, rest 60 s Sprint: 10 sprints on 30 m, rest 120 s
2	Lower body: 5 sets of 6 reps, RIR1, rest 150 s Upper body: 5 sets of 5 reps, RIR2, rest 60 s Sprint: 5 sprints on 20 m + 5 sprints on 30 m, rest 120 s
3	Lower body: 5 sets of 5 reps, RIR1, rest 150 s Upper body: 5 sets of 5 reps, RIR2, rest 60 s Sprint: 10 sprints on 20 m + 2 sprints on 30 m, rest 120 s
4	Lower body: 5 sets of 4 reps, RIR1, rest 150 s Upper body: 5 sets of 5 reps, RIR2, rest 60 s Sprint: 8 sprints on 30 m, rest 180 s
5	Lower body: 5, 4, 3, 2, 3, 4 reps (5 sets), RIR1, rest 180 s Upper body: 5 sets of 5 reps, RIR2, rest 60 s Sprint: 6 sprints on 20 m + 6 sprints on 30 m, rest 120 s
6	Lower body: 3 sets of 3 reps, RIR1 +5 reps (MEI, RIR3), rest 120 s Upper body: 5 sets of 5 reps, RIR2, rest 60 s Sprint: 6 sprints on 30 m, rest 180 s

HT: horizontal; VT: vertical; RIR: repetitions in reserve; MEI: maximum explosive intent.

cm-height box, performed at maximum explosive intent), while HT included barbell hip thrust, reverse hyperextension and horizontal jumps performed at maximum explosive intent. In both groups, the resistance used for each exercise was selected using the repetitions in reserve (RIR) method, and loads were chosen to avoid volitional failure in the sets.[14] Both training programs included the same total number of repetitions. The 6-week training program was characterized by a gradual decrease in training volume with a corresponding increase in training intensity. Jump exercises (vertical or horizontal) were performed at the beginning of each training session. Once a week, both groups performed the same sprint workouts that can be observed in Table II. The training sessions were supervised by the study investigators, and a standardized warm-up was performed before each workout and assessment session.

Performance assessments

Prior to performance assessments, participants performed a standardized warm-up consisting of five min on a cycle ergometer against a light resistance, 10 body weight squats, 10 body weight walking lunges, 10 dynamic walking hamstring stretches, and 10 dynamic walking quadriceps stretches.[15] The 1-RM test for the barbell parallel back squat and for the hip thrust was performed using methods previously described by Hoffman.[16] Briefly, each participant performed two warm-up sets using a resistance of approximately 40-60% and 60-80% of his perceived maximum, respectively. For each exercise, 3-4 subsequent trials were performed to determine the 1-RM. A 3-5 min rest period was provided between each trial. The 1RM assessments were performed under the supervision of a qualified investigator and trials not meeting the criteria set for range of motion and technique were discarded.

The countermovement jump (CMJ) test was performed using photoelectric cells (Optojump, Microgate, Bolzano, Italy). Subjects were instructed to maximize the height of each jump and self-select the countermovement depth, while keeping their hands on their hips. Flight time was calculated as the time interval from toe off to landing. Subjects performed 2 jumps with a 2-minute rest between each jump. The intrasession ICC calculated for the CMJ in this study was 0.93 (SEM: 1.93 cm; minimal difference: MD: 5.4 cm).

The standing long jump (SLJ) test was performed on a hard surface. Participants started from a standing position, with both tiptoes touching a starting line, and were allowed to swing their arms before the jump. Both the takeoff and landing phases of the jump had to be done with both feet. The distance between takeoff and the heel of the closest foot at landing was recorded in centimeters. Each participant completed two trials with a 2-min rest, and the best score was recorded and used for subsequent

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analysis.[17] The intrasession ICC calculated for the SLJ in this study was 0.94 (SEM: 3.95 cm; minimal difference: MD: 6.1 cm).

The 20 meters sprint test (20m SPRINT) was performed on an indoor court. Two pairs of photocells (Witty Gate, Microgate, Bolzano, Italy), set at 0 and 20 meters, were used to record the performance. Players start each sprint from a standing start with their front foot 0.5 m behind the first timing gate.[18, 19] Participants performed two trials with 3-min rest between each run. The ICC calculated for the 20-m sprint in this study was 0.89 (SEM: 0.16 s; minimal difference: MD: 0.11).

Ultrasonography measurements and body composition

Noninvasive skeletal muscle ultrasound images were collected from the participant's right side. Before image collection, all anatomical locations of interest were identified using standardized landmarks for the vastus lateralis muscle (VL), the vastus medialis muscle (VM), and the superior part of gluteus muscle (GLUT). The VL muscle thickness (MT) was measured along its longitudinal distance at 50% from the lateral condyle of the tibia to the most prominent point of the great trochanter of the femur, with the knee bent 10°.[20] The landmark for VM was positioned on the muscle belly at 22% of the distance between the upper edge of the patella and the superior iliac spine.[21] The landmark for the GLUT was identified on the lower half of a line which connected the tip of the greater trochanter to the anterior quarter of a line which directly connected the anterior and posterior iliac spines.[22] Participants were asked to lie on the examination table for at least 15 minutes before the images were taken. The same investigator performed all landmark measurements for each participant.

A 12 MHz linear probe scanning head (Echo Wave 2, Telemed Ultrasound Medical System, Milan, Italy) was coated with water soluble transmission gel to optimize spatial resolution and used to collect all ultrasound images. The probe was positioned on the surface of the skin without depressing the dermal layer and the view mode (gain: 50 dB; image depth: 5 cm) was used to take panoramic pictures of the VL. All ultrasound images were taken and analyzed by the same technician. MT measures were obtained using a longitudinal B-mode image. Three consecutive MT images were captured and analyzed for each muscle. For each image, MT was measured with a single perpendicular line from the superficial aponeurosis to the deep aponeurosis. The average of the three MT measures was used for statistical analyses. Intraclass correlation coefficients, 0.97 (SEM=0.55), and 0.95 (SEM=0.88) for, VLMT, and VMMT respectively. Echo intensity (EI) was assessed by computer-aided grey-scale analysis using ImageJ (National Institute of Health, Bethesda, MD, USA, Version 1.45) and was calculated for VL, VM and GLUT. The raw EI values were determined for the VL, VM and GLUT muscles as the corresponding index of muscle quality ranging between 0 and 255 a.u. (black: 0; white: 255). Anthropometric evaluations were performed before and after the training period. Body measurements included body mass, height, and body fat. Body mass was measured to the nearest 0.1 kg (Seca 769, Seca Scale Corp., Munich, Germany). Body fat percentage was estimated from skinfold caliper (Harpender; CMS Instruments, London, UK) measures using previously published methods.[23]

Statistical analysis

A Shapiro-Wilk Test was used to test the normal distribution of the data. If the assumption of sphericity was violated, a Greenhouse Geisser correction was applied. The differences in performance parameters, muscle architecture and body composition were calculated using a group (HT and VT) × time (pre – post) repeated measures analysis of variance (ANOVA). Pairwise comparisons were performed using the Bonferroni's *post-hoc*. Percent changes were calculated as follows: [(post-exercise mean – pre-exercise mean)/pre-exercise mean] × 100. The partial η^2 statistic was reported as the effect size (ES), and according to Stevens,[24] 0.01, 0.06, and 0.14 represent small, medium, and large effect sizes, respectively. Pearson product moment correlations were used to examine selected bivariate relationships. Significance was accepted at an alpha level of $P \leq 0.05$, and all data are reported as mean±SD. All data were analyzed using SPSS20 for Windows (SPSS Inc., Chicago, IL, USA).

Results

Performance assessments

Percentage changes in performance following the HT and VT programs are reported in Figure 1 and Table III. Significant group × time interactions were detected for 1RM squat (F=9.937; P=0.007; η^2=0.998), 1RM hip thrust (F=11.257; P=0.004; η^2=0.413) and standing long jump (F=6.048; P=0.026; η^2=0.274). The improvement in 1RM squat was more elevated in VT compared to HT (10.5% and 0.1%, respectively). Greater improvements were obtained in HT compared to VT for 1RM hip thrust (+17.9% and +5.3%, respectively), and for standing long jump (+7.6% and +1.6% in HT and VT, respectively). No sig-

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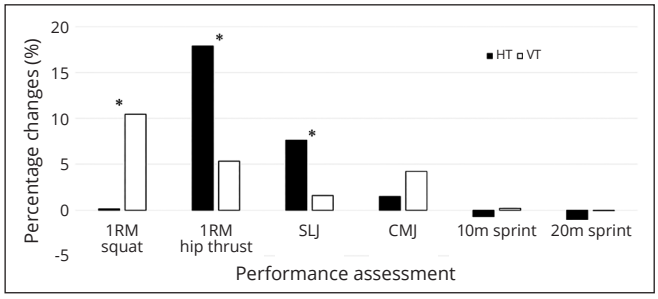

Figure 1.—Percentage changes in performance assessments in the HT and VT groups.
1RM: 1 repetition maximum; HT: horizontal; VT: vertical; SLJ: standing long jump; CMJ: counter movement jump.
*Significant (P≤0.05) groups × time interaction.

TABLE III.—*Performance parameters evaluated pre and post the training period in both HT and VT groups.*

Assessment	Timepoint	Group	
		HT	VT
1RM squat (kg)*	Pre	106.06±44.27	114.27±29.43
	Post	105.70±43.48	124.56±26.74
1RM hip thrust (kg)*	Pre	171.48±41.24	183.89±38.69
	Post	207.56±65.01	187.28±40.16
SLJ (cm)*	Pre	224.70±34.36	235.78±19.92
	Post	240.44±37.06	238.17±14.34
CMJ (cm)	Pre	39.10±8.58	39.40±8.13
	Post	39.65±8.72	40.67±6.90
20-m sprint (s)	Pre	3.45±0.30	3.31±0.14
	Post	3.41±0.25	3.31±0.12

HT: horizontal; VT: vertical; 1RM: 1 repetition maximum; CMJ: counter-movement jump; SLJ: standing long jump.
*Significant (P≤0.05) groups × time interaction.

nificant group × time interactions were detected for CMJ and 20-m sprint (P>0.05).

A significant main effect of time was observed for CMJ (F=4.233; P=0.047; η^2=0.220), 1RM squat (F=8.692; P=0.010; η^2=0.367), 1RM hip thrust (F=16.397; P=0.001; η^2=0.506) and standing long jump (F=11.183; P=0.004; η^2=0.41). No other significant main effects of time were detected (P>0.05).

Muscle architecture and body composition

Changes in body composition and muscle architecture following both training programs are reported in Table IV. In addition, percentage changes in muscle architecture are depicted in Figure 2. A significant group × time interaction was identified for VMMT (F=8.695; P=0.009; η^2=0.338). An increase of 4.1% was registered in VT, while a reduction by 7.9% was observed in HT. No group × time interactions were detected for VLMT, GLUTMT, VMEI, VLEI, GLUTEI (P>0.005).

TABLE IV.—*Anthropometric and muscle architecture measurements collected pre and post the training period in HT and VT group.*

Assessment	Timepoint	Group	
		HT	VT
Body mass (kg)	Pre	70.46±11.66	76.20±10.76
	Post	70.52±11.57	75.73±10.39
Fat mass (%)	Pre	14.99±7.01	11.87±3.62
	Post	14.40±6.09	11.48±3.11
VLMT (mm)	Pre	1.46±0.34	1.63±0.37
	Post	1.41±0.29	1.62±0.30
VMMT (mm)*	Pre	3.22±0.70	2.98±0.24
	Post	2.93±0.50	3.10±0.37
GLUTMT (mm)	Pre	3.70±1.09	4.10±0.73
	Post	4.22±1.22	4.21±0.49
VLEI (a.u.)	Pre	47.86±15.12	46.07±5.60
	Post	50.50±18.08	46.20±6.76
VMEI (a.u.)	Pre	32.71±7.57	38.18±6.80
	Post	33.55±7.68	36.11±4.67
GLUTEI (a.u.)	Pre	34.08±9.17	31.06±5.83
	Post	28.09±5.97	29.87±5.10

HT: horizontal; VT: vertical; VLMT: vastus lateralis muscle thickness; VMMT: vastus medialis muscle thickness, GLUTMT: gluteus muscle thickness; VLEI: vastus lateralis echo intensity; VMEI: vastus medialis echo intensity; GLUTEI: gluteus echo intensity.
*Significant (P≤0.05) groups × time interaction.

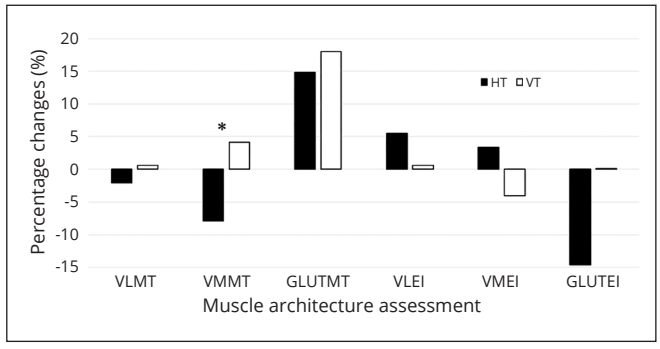

Figure 2.—Percentage changes in muscle architecture and echo intensity following the HT and VT programs.
HT: horizontal; VT: vertical; VL: vastus lateralis muscle; VM: vastus medialis muscle; GLUT: gluteus maximus muscle; MT: muscle thickness; EI: echo intensity.
*Significant (P≤0.05) groups × time interaction.

Significant main effects for the factor time were observed for GLUTMT (F=30.758; P=0.000; η^2=0.658) and GLUTEI (F=10.946; P=0.050; η^2=0.412). No other main effects of time were detected.

Correlations between variables

Significant correlations were detected between VMMT and 1 RM squat (*r*=0.605; P=0,010) and between VMMT and

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1RM hip thrust (r=0.577; P=0.012). GLUTMT was positive correlated to CMJ (r=0.649; P=0.004), SLJ (r=0.657; P=0.002), 1 RM squat (r=0.617; P=0.008), and 1RM hip thrust (r=0.769; P<0.001). GLUTEI was negatively correlated to CMJ (r=-0.487; P=0.004), and SLJ (r=-0.685; P=0.001). In addition, GLUTEI and VLEI were correlated with 20m SPRINT (r=0.792; P<0.001 and r=0.69; P<0.001 for GLUTEI and VLEI respectively). No other significant correlations were registered.

Discussion

This study was conceived to compare the effects of a horizontally *vs.* a vertically based training program on maximal strength, performance and muscle architecture of the lower body muscles. According to the research hypotheses, the VT program was more effective than the HT program in improving 1RM squat while the HT group increased more the 1RM hip thrust. These results confirm the principle of specificity since squat and hip thrust were included solely in the HT and VT program, respectively. Curiously, while the HT group experienced a more relevant increase in standing long jump performance compared to the VT group, both groups obtained a similar improvement in vertical jump. Some authors[25] reported that vertical jump performance was more dependent by the power of hip extensors muscles compared to knee extensors. Since, both vertical and horizontal exercises activate the hip extensor muscles and, in the present study, both groups obtained the same increase in MT (+14.8% and +18.0% for HT and VT, respectively), vertically and horizontally oriented exercises may produce similar effects on vertical jump performance. Consistently with the aforementioned principle of specificity however, a greater increase in standing long jump performance was registered in HT compared to VT group. Although some authors reported greater improvements in on 0-10 m and 0-20 m sprint performances following 6 weeks of hip thrust compared to front squat,[11] in the present study neither group obtained significant improvements in sprint. Since participants were resistance trained adults, the duration of the training period may not be enough to stimulate improvements in this performance. Conversely, in the study by Contreras *et al.*[11] participants were adolescent athletes that may be more prone to increase their sprint performance compared to trained adults.

In the present study, greater gains in MT of the VM were detected in VT compared to HT group. Vertically loaded exercises such as back squats are known to induce higher excitations of the quadriceps muscles compared to horizontally loaded exercises (*e.g.*, hip thrust).[26] In addition, the vertically loaded resistance exercises included in the VT program (squat, step up), represent multi joint exercises involving both hip and knee extensor muscles. On the contrary, horizontally loaded exercises included in HT program (hip thrust and reverse hyperextension) are single joint movements mainly involving the hip extensor muscles. Thus, knee extensor muscles such as VM were less stimulated by HT compared to VT program.

Muscle EI has been indicated as marker of muscle quality and changes in this parameter have been reported by some authors following resistance training in old[27] and young individuals.[28] Other authors, however, were not able to find relevant exercise-induced changes in EI.[29] In the present study, a reduction in the echo intensity of the GLUT (indicating a possible increase in muscle quality) was detected in both groups following the training period. This is consistent with the similar increase in the muscle thickness of GLUT noted in both groups. Thus, both VT and HT programs were equally effective in stimulating muscle growth and enhancing muscle quality of the GLUT. Although very high levels of activation of the gluteus are typically associated to horizontally loaded exercises[30] (Hip thrust and hyperextension), vertically loaded exercises may induce similar muscle activations.[31] In particular, the step-up exercise, performed at high loads, presents a massive activation of the gluteus muscle.[10] To the best of our knowledge, this is the first study to include the reverse-hyperextension into a resistance training program. Despite the high level of activation registered in the muscles of the posterior chain during this movement,[8] the HT group did not show superior gain in muscle size and echo intensity of the gluteus compared to the VT group. In addition, no changes in the echo intensity of knee extensor muscles were registered in both groups.

Limitations of the study

The large correlation between the echo intensity of both VL and GLUT and the performance in 20m SPRINT (r=0.69 and r=0.79 for VL and GLUT, respectively) are consistent with previous studies[32] showing how this parameter represents a key component for sprint. A possible limitation of the present study is represented by the short duration of the training period (6 weeks). Although this represents a common limitation of training studies, longer periods of training may induce further adaptations and a more realistic comparison between VT and HT.

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Conclusions

The results of the present study indicated that a 6-week resistance training program based on horizontally loaded exercises (HT) was more effective than a resistance program based on vertically loaded exercises (VT) for the improvement of horizontally-oriented tasks such as standing long jump and 1RM hip thrust but was equally effective as VT for vertical jump improvement. Both HT and VT programs were equally effective in promoting muscle growth and echo intensity of the GLUT, while VT was more effective than HT in maximizing the muscle growth of the knee extensors. Both horizontally and vertically loaded exercises may be successfully included, or combined, in resistance training programs to enhance performance and muscle hypertrophy of the lower body.

References

1. Mangine GT, Hoffman JR, Gonzalez AM, Townsend JR, Wells AJ, Jajtner AR, *et al.* The effect of training volume and intensity on improvements in muscular strength and size in resistance-trained men. Physiol Rep 2015;3:e12472.

2. Contreras B. The top five glute exercise; 2010 [Internet]. Available from: https://charlieweingroff.com/ [cited 2024, Oct 4].

3. Loturco I, Contreras B, Kobal R, Fernandes V, Moura N, Siqueira F, *et al.* Vertically and horizontally directed muscle power exercises: relationships with top-level sprint performance. PLoS One 2018;13:e0201475.

4. Fatouros IG, *et al.* Evaluation of Plyometric Exercise Training, Weight Training, and Their Combination on Vertical Jumping Performance and Leg Strength. J Strength Cond Res 2000;14:470.

5. Santos EJ, Janeira MA. Effects of complex training on explosive strength in adolescent male basketball players. J Strength Cond Res 2008;22:903–9.

6. Abade E, Sampaio J, Santos L, Gonçalves B, Sá P, Carvalho A, *et al.* Effects of using compound or complex strength-power training during in-season in team sports. Res Sports Med 2020;28:371–82.

7. Lawrence MA, Chin A, Swanson BT. Biomechanical Comparison of the Reverse Hyperextension Machine and the Hyperextension Exercise. J Strength Cond Res 2019;33:2053–6.

8. Cuthbert M, Ripley NJ, Suchomel TJ, Alejo R, McMahon JJ, Comfort P. Electromyographical Differences Between the Hyperextension and Reverse-Hyperextension. J Strength Cond Res 2021;35:1477–83.

9. Appleby BB, Newton RU, Cormack SJ. Kinetics and Kinematics of the Squat and Step-up in Well-Trained Rugby Players. J Strength Cond Res 2019;33(Suppl 1):S36–44.

10. Neto WK, Soares EG, Vieira TL, Aguiar R, Chola TA, Sampaio VL, *et al.* Gluteus Maximus Activation during Common Strength and Hypertrophy Exercises: A Systematic Review. J Sports Sci Med 2020;19:195–203.

11. Contreras B, Vigotsky AD, Schoenfeld BJ, Beardsley C, McMaster DT, Reyneke JH, *et al.* Effects of a Six-Week Hip Thrust vs. Front Squat Resistance Training Program on Performance in Adolescent Males: A Randomized Controlled Trial. J Strength Cond Res 2017;31:999–1008.

12. Barrera-Domínguez FJ, Martínez-García D, Jerez-Mayorga D, Chirosa-Ríos LJ, Almagro BJ, Molina-López J. Vertical Versus Horizontal Training for Improving the Change of Direction Speed in Adult Basketball Players: A Systematic Review and Meta-analysis. J Strength Cond Res 2023. [Epub ahead of print].

13. Bartolomei S, Malagoli Lanzoni I, Fantozzi S, Cortesi M. Comparison between Non-Localized Post-Activation Performance Enhancements Following Resistance Exercise for the Upper and the Lower Body. Adv Sci Inst Ser Appl Sci. 2022;12:1639.

14. Zourdos MC, Klemp A, Dolan C, Quiles JM, Schau KA, Jo E, *et al.* Novel Resistance Training-Specific Rating of Perceived Exertion Scale Measuring Repetitions in Reserve. J Strength Cond Res 2016;30:267–75.

15. Nigro F, Bartolomei S. A Comparison Between the Squat and the Deadlift for Lower Body Strength and Power Training. J Hum Kinet 2020;73:145–52.

16. Hoffman J. Physiological Aspects of Sport Training and Performance. Second Edition. Champaign, IL: Human Kinetics; 2014. p.520.

17. Saint-Maurice PF, Laurson KR, Kaj M, Csányi T. Establishing Normative Reference Values for Standing Broad Jump Among Hungarian Youth. Res Q Exerc Sport 2015;86(Suppl 1):S37–44.

18. Nikolaidis PT. Reference values for the sprint performance in male football players aged from 9–35 years. Biomed Hum Kinet 2016;8:103–12.

19. Grazioli R, Loturco I, Lopez P, Setuain I, Goulart J, Veeck F, *et al.* Effects of Moderate-to-Heavy Sled Training Using Different Magnitudes of Velocity Loss in Professional Soccer Players. J Strength Cond Res 2023;37:629–35.

20. Bemben MG. Use of diagnostic ultrasound for assessing muscle size. J Strength Cond Res 2002;16:103–8.

21. Blazevich AJ, Gill ND, Zhou S. Intra- and intermuscular variation in human quadriceps femoris architecture assessed in vivo. J Anat 2006;209:289–310.

22. Dieterich AV, Pickard CM, Strauss GR, Deshon LE, Gibson W, McKay J. Muscle thickness measurements to estimate gluteus medius and minimus activity levels. Man Ther 2014;19:453–60.

23. Evans EM, Rowe DA, Misic MM, Prior BM, Arngrímsson SA. Skinfold prediction equation for athletes developed using a four-component model. Med Sci Sports Exerc 2005;37:2006–11.

24. Stevens JP. Applied Multivariate Statistics for the Social Sciences. Fifth Edition. New York, NY: Routledge; 2012. p.664.

25. Marshall BM, Moran KA. Biomechanical Factors Associated With Jump Height: A Comparison of Cross-Sectional and Pre-to-Posttraining Change Findings. J Strength Cond Res 2015;29:3292–9.

26. Delgado J, Drinkwater EJ, Banyard HG, Haff GG, Nosaka K. Comparison Between Back Squat, Romanian Deadlift, and Barbell Hip Thrust for Leg and Hip Muscle Activities During Hip Extension. J Strength Cond Res 2019;33:2595–601.

27. Radaelli R, Botton CE, Wilhelm EN, Bottaro M, Lacerda F, Gaya A, *et al.* Low- and high-volume strength training induces similar neuromuscular improvements in muscle quality in elderly women. Exp Gerontol 2013;48:710–6.

28. Vasenina E, Kataoka R, Hammert WB, Ibrahim AH, Dankel SJ, Buckner SL. Examination of Changes in Echo Intensity Following Resistance Exercise among Various Regions of Interest. Clin Physiol Funct Imaging 2022;42:23–8.

29. Santos R, Valamatos MJ, Mil-Homens P, Armada-da-Silva PA. Muscle thickness and echo-intensity changes of the quadriceps femoris muscle during a strength training program. Radiography (Lond) 2018;24:e75–84.

30. Williams MJ, Gibson NV, Sorbie GG, Ugbolue UC, Brouner J, Easton C. Activation of the Gluteus Maximus During Performance of the Back Squat, Split Squat, and Barbell Hip Thrust and the Relationship With Maximal Sprinting. J Strength Cond Res 2021;35:16–24.

31. Macadam P, Feser EH. Examination Of Gluteus Maximus Electromyographic Excitation Associated With Dynamic Hip Extension During Body Weight Exercise: A Systematic Review. Int J Sports Phys Ther 2019;14:14–31.

32. Bartolomei S. Influences of Sex on Muscle Architecture and Performance in Elite Field Hockey Players. NATO Adv Sci Inst Ser E Appl Sci. 2023;13:11314.

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Conflicts of interest
The authors certify that there is no conflict of interest with any financial organization regarding the material discussed in the manuscript.

Authors' contributions
Sandro Bartolomei has given substantial contributions to study conception, Sandro Bartolomei and Giuseppe Rosaci to study design, data collection and manuscript writing, Davide Latini and Federico Nigro to manuscript writing and data analysis. All authors read and approved the final version of the manuscript.

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